

Supermarket Sales Analysis Project Report

Project Title: Supermarket Sales Analysis and Dashboard

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1. Introduction

This project focuses on analyzing the sales data of a supermarket chain to generate meaningful insights for business decision-making. The goal was to clean the raw data, perform exploratory data analysis using Excel, and build an interactive dashboard in Power BI.

2. Data Cleaning

Before analysis, the raw dataset had several issues:

- Duplicate Entries: Removed identical rows.
- Missing Values: Replaced nulls in text columns with "Unknown" and numeric columns with 0.
- Inconsistent Date Formats: Standardized all dates to YYYY-MM-DD.
- Untrimmed/Misformatted Text: Trimmed whitespaces and unified casing in key text fields.

These steps ensured the data was accurate and usable for further analysis.

3. Exploratory Analysis with Excel

I used Pivot Tables in Excel to explore the cleaned data and calculate the following KPIs:

- Total Sales
- Total Profit
- Sales Quantity
- Average Order Value
- Profit by State
- Top-Selling Products
- Sales by Region
- Sales by Month
- Top Cities by Sales

These KPIs were first validated in Excel to ensure correct data trends.

4. Power BI Dashboard

Using the same KPIs, I developed a Power BI dashboard with the following components:

- Cards displaying Total Sales, Profit, and Quantity Sold.
- Profit by State: A bar chart identifying high and low-performing states.
- Top-Selling Products: Products that contributed most to revenue.
- Sales by Month: A line chart showing seasonal trends.
- Sales by Region: A breakdown of regional performance.
- Top Cities by Sales: A chart showing which cities bring in the most revenue.

This dashboard helps stakeholders understand business performance at a glance.

5. Insights & Takeaways

From the analysis:

- West Region leads in total sales.
- Technology is the most profitable category.
- Certain months show spikes in sales indicating seasonality.
- Top cities like New York and Los Angeles generate the highest revenue.
- Some high-selling products have low profit margins, needing pricing review.

6. Tools Used

- Microsoft Excel: Data cleaning, pivot tables, and initial KPI validation.
- Microsoft Power BI: Interactive dashboard and data visualization.
- Power Query: For data transformation and date formatting.

7. Conclusion

This project gave me hands-on experience in:

- Data Cleaning
- KPI calculation using Excel
- Building Dashboards in Power BI
- Extracting insights from structured data

It demonstrates how data can be turned into actionable decisions using the right tools and techniques.

8. Appendix

- Cleaned Dataset: Superstore_cleaned.csv
- Dirty Dataset for Demo: Superstore_unclean.csv
- Power BI Dashboard: sale dashboard of supermarket.pbix
- screenshot of dashborad